

The Illusion of 5 Stars: Cross-City Rating Calibration and Visibility Disparities on Airbnb

Xinyue (Shirley) Zhang¹

¹Department of Computer Science, Stanford University



Overview

Airbnb ratings are often treated as universal measures of quality. However, prior work suggests that Airbnb ratings are highly inflated and shaped by social context (1; 2). Using Inside Airbnb data from Berlin, Chicago, Hong Kong, and Tokyo, we examine whether identical ratings carry equivalent meanings across cities and whether calibration differences affect platform rankings. This project examines how local evaluation norms become embedded within platform reputation systems (3). When rankings treat ratings as directly comparable across markets, seemingly small differences in calibration can translate into meaningful differences in visibility and opportunity.

Methods

We evaluate rating calibration through five complementary analyses steps that progress from descriptive patterns to downstream platform consequences:

- Rating Distributions:** Compared rating histograms, ECDFs, and threshold pass rates to determine whether common ratings (e.g., 4.8 and 4.9 stars) correspond to similar positions within local rating distributions.
- Comparable Listing Cohorts:** Compared ratings among listings with similar room types, occupancy sizes, review volumes, and price ranges to reduce differences in housing inventory.
- Review Language Analysis:** Used language detection, VADER sentiment analysis (4), and multilingual style classification to examine whether reviews associated with similar ratings differed across cities.
- Regression Analysis:** Estimated OLS models controlling for price, capacity, review volume, room type, Superhost status, and property type to isolate city-level rating effects.
- Ranking Simulation:** Compared raw Airbnb rankings with city-normalized rankings based on local rating percentiles to evaluate how calibration differences affect visibility.

Experiment 1: Rating Distributions

Hypothesis. If ratings are not equally calibrated across cities, common rating thresholds should represent different proportions of listings across markets.

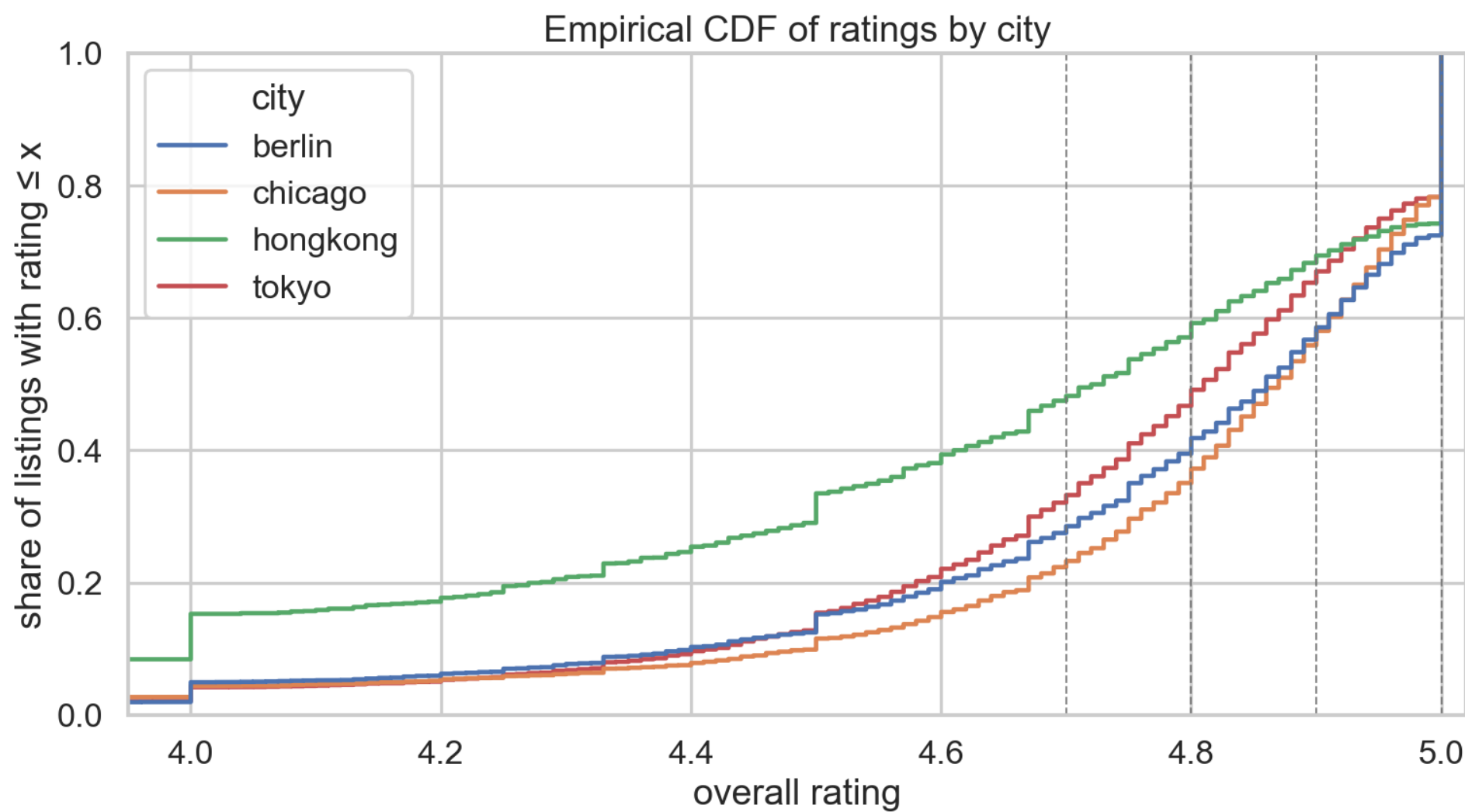


Figure 1. *

Empirical CDFs of Airbnb ratings by city. Vertical lines indicate common rating thresholds.

Chicago and Berlin have larger shares of listings above common thresholds such as 4.8 and 4.9 stars, while Hong Kong has substantially fewer listings above these cutoffs. As a result, identical ratings correspond to different relative standings across cities, creating the potential for unequal visibility when rankings rely directly on raw star ratings.

Experiment 2: Comparable Listing Cohorts

Hypothesis. If ratings are calibrated differently across cities, cross-city rating differences should remain visible even among broadly comparable listings.

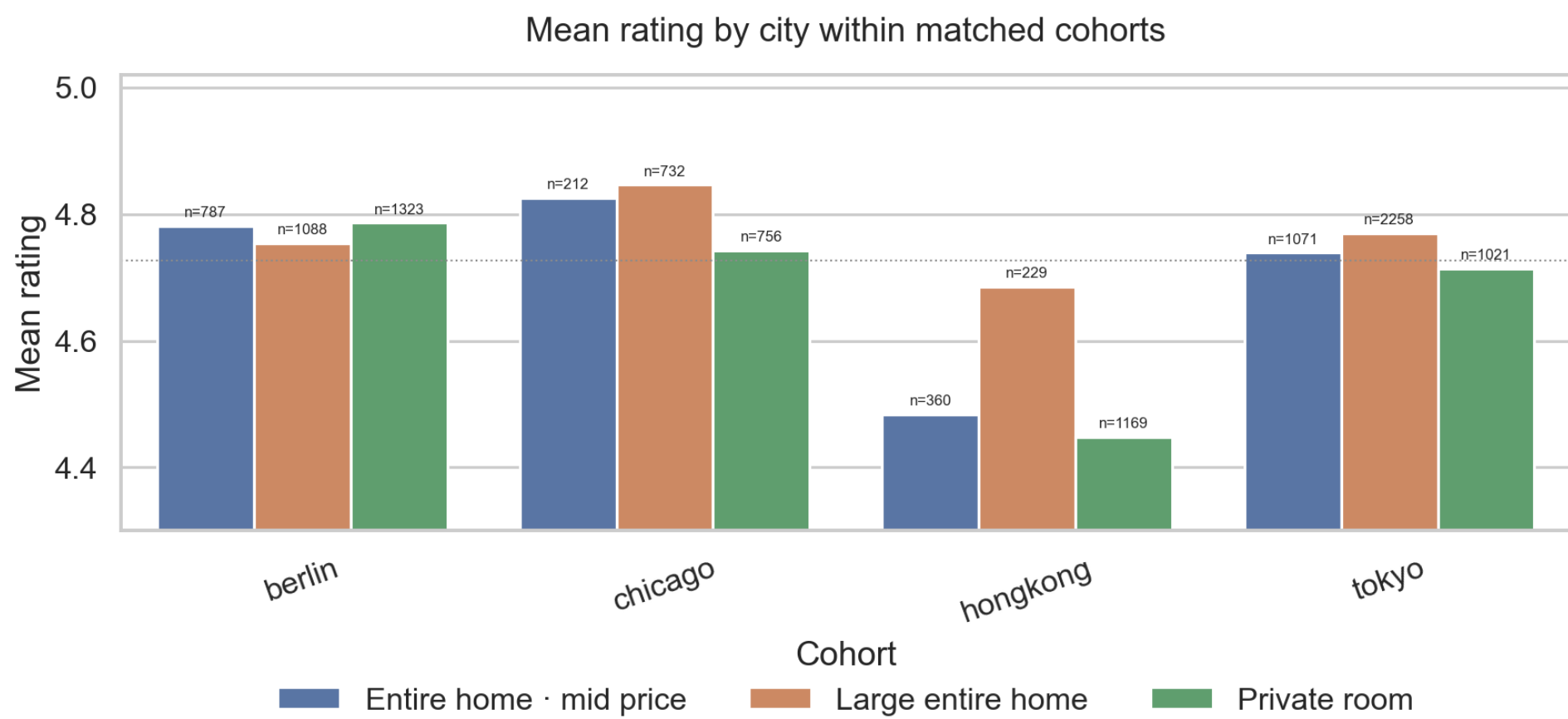


Figure 2. *

Mean ratings within matched cohorts: mid-price homes, large homes, and private rooms.

Listings were compared within three matched cohorts to reduce differences in housing inventory. Hong Kong remained the lowest-rated city across cohorts, while Chicago and Berlin generally remained near the top. These persistent gaps suggest that cross-city rating differences reflect local evaluation norms in addition to listing characteristics.

Experiments 3–4: Text and Regression Evidence

Review Language

Reviews with similar ratings exhibited different linguistic patterns across cities. Chicago reviews were generally more polite and enthusiastic, while Hong Kong reviews were less expressive, even among English-language reviews.

Regression Controls

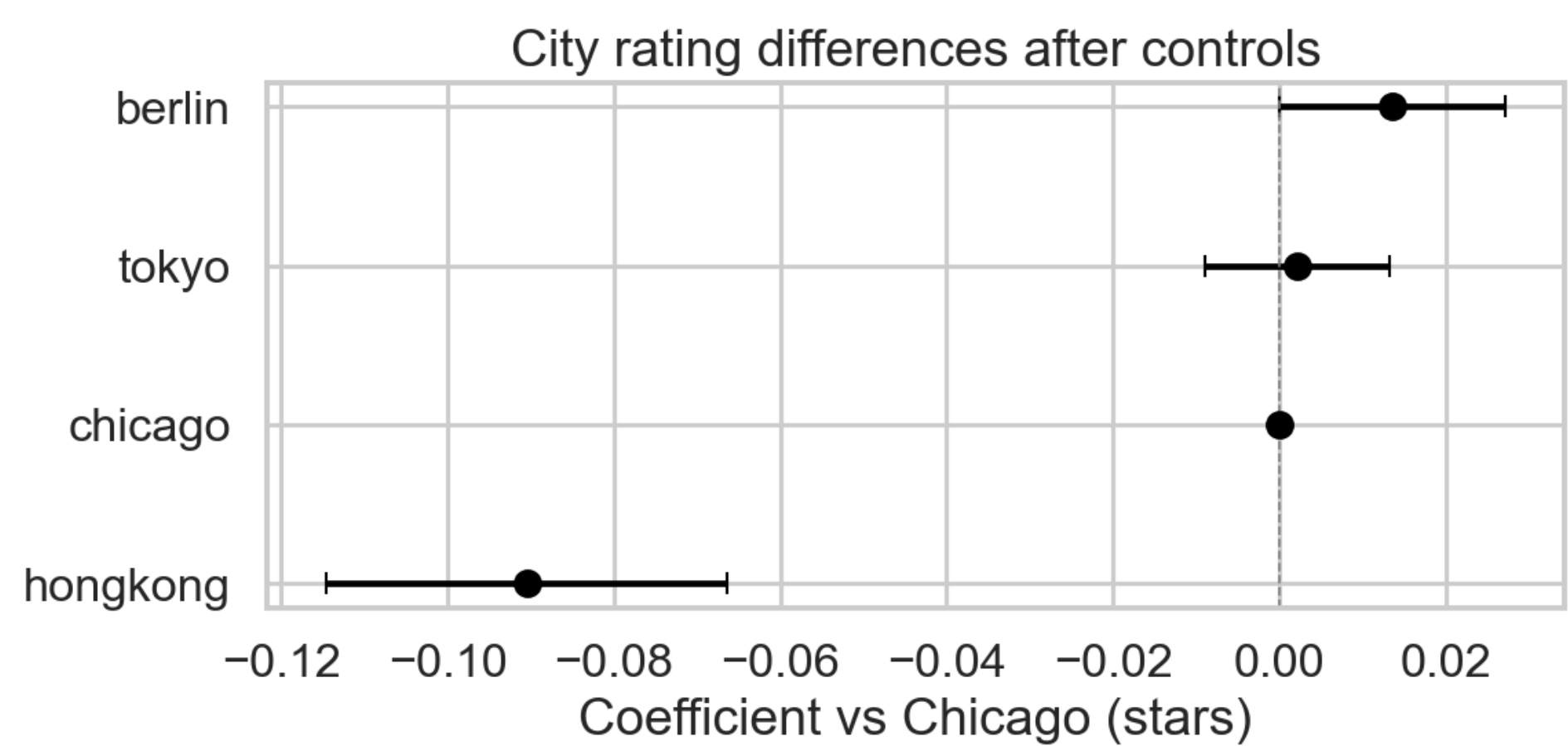


Figure 3. *

Estimated city effects relative to Chicago after controlling for listing characteristics. Error bars show 95% confidence intervals.

OLS models controlled for price, capacity, review volume, room type, Superhost status, and property type. Hong Kong remained significantly lower-rated after controls, while Berlin remained slightly higher and Tokyo remained similar to Chicago, suggesting that listing characteristics alone cannot explain cross-city rating differences.

Experiment 5: Ranking Simulation

Hypothesis. Small calibration differences can produce larger differences in visibility when rankings are based on raw star ratings. Listings were ranked under two systems:

- Raw ranking:** listings ranked directly by Airbnb star rating.
- City-normalized ranking:** listings ranked by empirical percentile within each city's ratings.

The normalized ranking is not assumed to be objectively superior; it is used as a counterfactual to measure how calibration differences affect ranking outcomes.

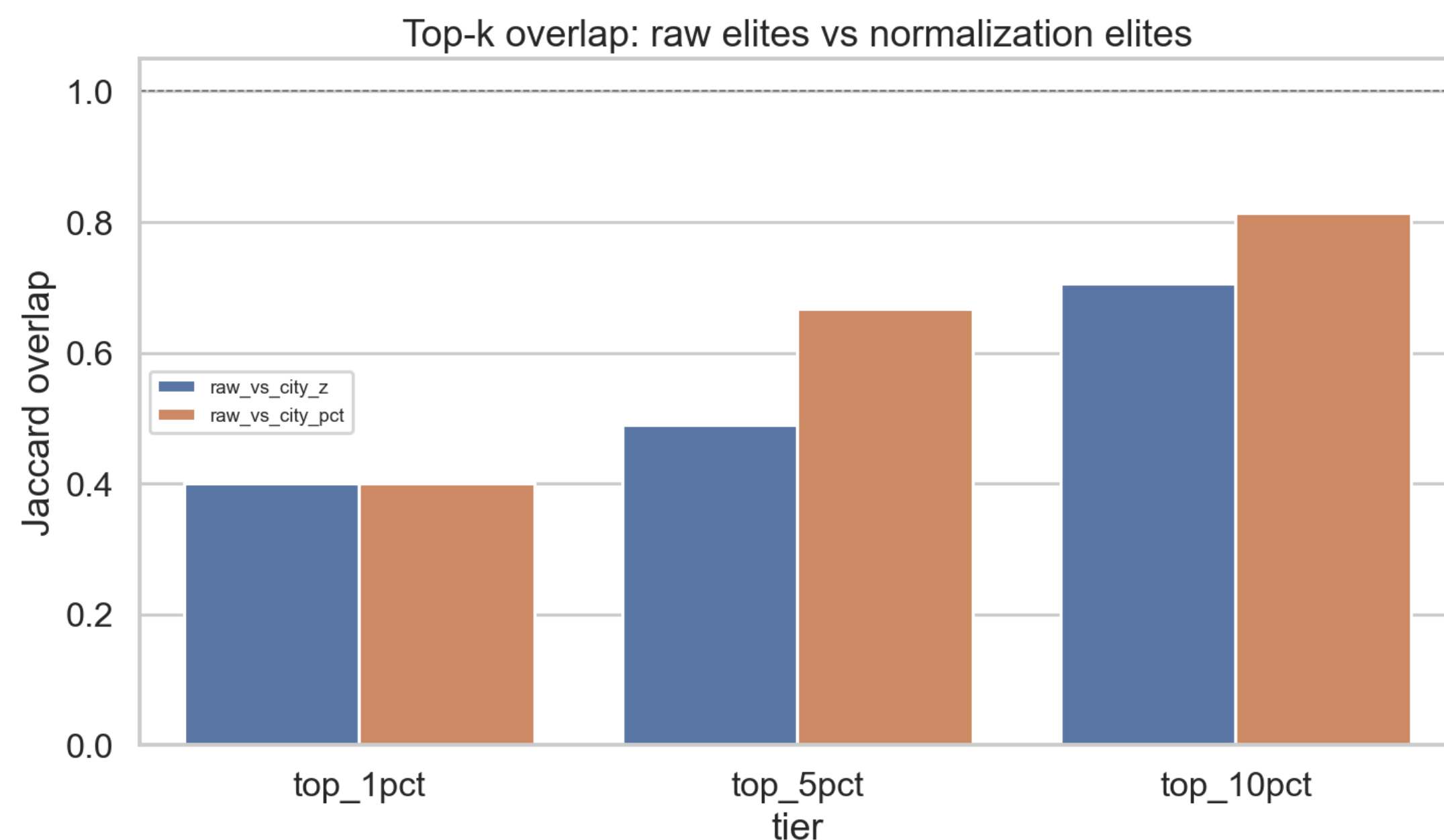


Figure 4. *

Overlap between top-ranked listings under raw rating and city-normalized rankings.

Only about 40% of listings appearing in the top 1% under raw ratings remained in the top 1% after normalization. Even within the top 10%, overlap reached only about 70–80%, indicating that many highly ranked listings changed when local rating distributions were taken into account.

Conclusion and Implications

Across multiple analyses, we find that Airbnb ratings are not calibrated uniformly across cities. Differences in rating distributions persisted among comparable listings and after controlling for listing characteristics, suggesting that ratings may reflect local evaluation norms or other market-level factors in addition to service quality.

Because platform ratings are used by ranking, search, and recommendation systems, even small calibration differences can be amplified into meaningful differences in visibility. The findings challenge the assumption that a standardized five-star scale represents a universal measure of quality. When platforms rank listings using globally comparable ratings, local evaluation norms can translate into substantial differences in exposure and opportunity across markets.

References

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